LESSON PLAN

PISTON ENGINE (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Description & operation of simple float type carburetor, modified carburetor, limitation Types of Engine starters, & trouble shooting on piston engine
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Description & operation of simple float type carburetor, modified carburetor, limitation Types of Engine starters, & trouble shooting on piston engine
5. Specific Objectives	15.1.4.1 Trainees will be able to explain operation of carburetor.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of carburetor float level adjustment and choke operation; Instructor supervision required during model handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a carburetor in a piston engine?
(b) Name two types of engine starters used in aircraft.
(c) List three common symptoms of a carburetor malfunction.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Simple Float Type Carburetor Description • Operation of Simple Float Carburetor • Modified Carburetor Description & Operation • Limitations of Simple Float Carburetor	L & Ds	PPT, IB	15 min
(b)	• Types of Engine Starters • Electric Starter System Operation • Air Starter System Operation • Hydraulic Starter System Overview	L & Ds	PPT, PDF	20 min
(c)	• Troubleshooting Carburetor Issues • Diagnosis of Fuel Flow Problems • Common Carburetor Malfunctions • Corrective Actions for Engine Roughness	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the float level affects the fuel-air mixture in a simple carburetor.
(b) Differentiate between the operation of an electric starter and an air starter.
(c) Identify two limitations of the simple float type carburetor compared to the modified type.

12. Home Assignment and Follow Up	Prepare a brief report comparing the operational principles of the simple float carburetor and the modified carburetor, highlighting key differences relevant to future engine maintenance tasks in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min